

A. caulescens. Clump-forming perennial. H 3–4ft (90–120cm), S 2ft (60cm). From midsummer to early fall, leeklike stems carry open, rounded umbels of bell-shaped violet-blue flowers. Lower leaves are narrowly strap-shaped; upper leaves are broader. Z6–10

A. inapertus. Clump-forming perennial. H 3–5ft (90cm–1.5m), S 2ft (60cm). Pendent, narrowly tubular, blue flowers are borne on very erect stems, above narrow, bluish-green leaves, in late summer and fall. Z6–10.

subsp. pendulus ‘Graskop’ illus. p.240.

A. ‘Lilliput’. Compact, clump-forming perennial. H and S 16in (40cm). Has small, rounded umbels of dark blue flowers that are produced in summer. Leaves are narrow and mid-green. Z6–9

A. ‘Loch Hope’. Clump-forming perennial. H to 5ft (1.5m), S 2ft (60cm). Bears large, rounded umbels of deep blue flowers in late summer and early fall, above narrow, grayish-green leaves. Z8–11

A. ‘Northern Star’ illus. p.241.

A. ‘Phantom’. Evergreen, clump-forming perennial. H 4ft (1.2m), S 2ft (60cm) or more. Large, rounded clusters of white flowers, flushed pale blue at the margins, are borne on sturdy stems in mid- to late summer. Has broadly strap-shaped, bright green leaves. Z9–11

A. praecox. Variable, evergreen perennial. H to 4ft (1.2m), S 18in (45cm). Has rounded umbels of usually trumpet-shaped, white or blue flowers in late summer and early fall. Z8–10. ‘**Albiflorus**’ is white-flowered.

A. ‘Purple Cloud’ illus. p.241.

AGAPETES

SYN. PENTAPTERYGIIUM

ERICACEAE

Genus of evergreen, occasionally briefly deciduous, spreading to erect scandent shrubs, grown for their flowers. Provide full light or partial shade and a humus-rich, well-drained but not dry, neutral to acidic soil. Water containerized specimens freely when in full growth, but moderately at other times. Overly long stems may be cut back to promote branching, but they are best tied to supports. Propagate by seed sown in spring or by semiripe cuttings taken in late summer.

A. incurvata, syn. *A. rugosa* var. *rugosa* of gardens. Evergreen, loose shrub, which may be trained as a climber, with arching to pendent stems. H to 3ft (1m), S 3–6ft (1–2m) as a shrub; H 6–10ft (2–3m) as a climber. Leaves are lance-shaped to ovate, wrinkled and bright green. In spring, bears clusters of pendent, urn-shaped, white flowers, patterned with purple-red, from leaf axils. Z12–13 (min. 41–15°F/5–7°C)

A. ‘Ludgvan Cross’. Evergreen, scandent shrub with arching or pendulous stems. H and S 4–5ft (1.2–1.5m). Produces lance-shaped, dark green leaves, and urn-shaped, red flowers, with darker patterns, in spring. Z10–11

A. macrantha. See *A. variegata* var. *macrantha*.

A. rugosa var. rugosa of gardens. See *A. incurvata*.

A. serpens illus. p.461.

A. variegata var. macrantha, syn. *A. macrantha*, illus. p.461.

AGASTACHE

Mexican giant hyssop

LAMIACEAE/LABIATAE

Genus of summer-flowering perennials with aromatic leaves. Requires full sun and fertile, well-drained soil. Plants are short-lived and should be propagated each year by softwood or semiripe cuttings taken in late summer.

A. ‘Black Adder’ illus. p.280.

A. mexicana, syn. *Brittonastrum mexicanum*, *Cedronella mexicana*. Bushy, upright perennial with aromatic leaves. H 2–3ft (60–90cm), S 1ft (30cm). Whorls of small, tubular flowers in shades of pink to crimson are borne in dense spikes in mid- and late summer. Leaves are ovate, pointed, toothed and mid-green. Z7–11

Agathaea. See *Felicia*.

AGATHOSMA

RUTACEAE

Genus of evergreen shrubs, grown for their flowers and overall appearance. Requires full light and well-drained, acidic soil. Water containerized plants moderately, less when not in full growth. Propagate by semiripe cuttings in late summer.

A. pulchella, syn. *Barosma pulchella*. Evergreen, rounded, wiry, aromatic shrub. H and S to 3ft (1m). Produces a dense mass of small, ovate, leathery leaves. Terminal clusters of small, 5-petaled, purple flowers are freely borne in spring–summer. Z9–11

AGAVE

ASPARAGACEAE/AGAVACEAE

Genus of rosette-forming, perennial or monocarpic succulents with sword-shaped, sharp-toothed leaves. Small species, to 1ft (30cm) high, flower only after 5–10 years; tall species, to 15ft (5m) high, may take 20–40 years to flower. The majority of species that have hard, blue-gray leaves are hardy; gray-green- or green-leaved species are usually frost-tender. Requires full sun and well-drained soil. Propagate by seed or offsets in spring or summer.

A. americana (Century plant). Basal-rosetted, perennial succulent. H 3–6ft (1–2m), S 6–10ft (2–3m). Has sharply pointed, toothed leaves, to 5–6ft (1.5–2m) long. Branched flower stem, to 25ft (8m) long, bears dense, tapering spikes of bell-shaped, white to pale creamy-yellow flowers, each 3½in (9cm) long, in spring–summer. Offsets freely. Z8–11. (min. 41°F/5°C). ‘**Marginata**’ (illus. p.482) has yellow margins, which often become white with age, and a central green zone to each leaf. ‘**Mediopicta**’ has central, yellow stripes along leaves. ‘**Striata**’ illus. p.481.

A. attenuata illus. p.489.

A. duplicata. See *Polianthes geminiflora*.

A. filifera (Thread agave; illus. p.482). Basal-rosetted, perennial succulent. H 12–20in (30–50cm), S 3ft (1m). Narrow, green leaves are spined at the tip. White leaf margins gradually break away, leaving long, white fibers. Produces yellow-green flowers on a 8ft (2.5m) tall stem in summer. Offsets freely. Z10–11 (min. 41°F/5°C)

A. geminiflora (illus. p.482). Basal-rosetted, perennial succulent. H to 2ft (60cm) or 18ft (6m) in flower, S to 3ft (1m). Has very narrow, linear, smooth green leaves, to 2ft (60cm) long, with fine filaments along the edges. In late summer, bears a tall, spike of up to 1,500 yellow-green to reddish flowers, to 2in (5cm) long. Z8–11

A. hystrix. See *A. stricta*.

A. macroacantha (illus. p.482). Basal-rosetted, perennial succulent. H 12–16in (30–40cm), S 10–16in (25–40cm). Narrowly sword-shaped, toothed, bluish-gray leaves have dark brown spines. Flower stem, to 6ft (2m) long, bears slender panicles of tubular, purple-tinged, green flowers, 2in (5cm) long, and sometimes bulbils, in summer. Z8–11 (min. 41°F/5°C)

A. maculosa, syn. *Manfreda maculosa*. Basal-rosetted, perennial succulent. H 24–72in (60cm–1.8m), S 12–36in (30–90cm). Lance-shaped, grooved, dark green leaves, with small, distantly spaced teeth, and sometimes dark green- or brown-spotted patterns. Flower stem, to 6ft (1.8m) long, bears narrow spikes of bell-shaped, white to yellowish-white flowers, ¾in (1.5cm) long, in summer. Z8–11 (min. 41°F/5°C)

A. parrasana (illus. p.482). Basal-rosetted, compact, perennial succulent. H 12–16in (30–40cm), S 12–18in (30–45cm). Ovate, closely packed and overlapping, thick, rigid, light gray leaves have toothed margins with short, grayish-brown spines. Flower stem, 10–13ft (3–4m) long, bears ellipsoidal panicles of bell-shaped, pale yellow flowers, 2in (5cm) long, flushed red or purple, in summer. Z9–11

A. parryi illus. p.489.

A. parviflora (illus. p.482). Basal-rosetted, perennial succulent. H 6in (15cm), S 20in (50cm). Has narrow, white-marked, dark green leaves with white fibers peeling from edges. Produces white or pale yellow flowers in summer. Z10–11 (min. 41°F/5°C)

A. polianthes. See *Polianthes tuberosa*.

A. scaphoidea. See *A. utahensis*.

A. schottii. Basal-rosetted, perennial succulent. H 8–16in (20–40cm), S 12–18in (30–45cm). Narrowly linear, yellowish green to green leaves have margins bearing brittle threads. Often crooked flower stem, to 6ft (2m) long, produces slender spike of tubular, yellow flowers, 1¼–1½in (3–4cm) long, in summer. Z8–11

A. stricta, syn. *A. hystrix* (illus. p.482). Short-stemmed, perennial succulent. H and S 10–20in (25–50cm). Produces rosettes of linear-lance-shaped, sharply tapered, mid-green leaves, to 14in (35cm) long, with red-brown terminal spines. In summer, bears red to purple-red flowers, ¾in (2cm) long, in dense, erect, spikelike racemes, 6–8ft (2–2.5m) long. Z8–10 (min. 41°F/5°C)

A. utahensis, syn. *A. scaphoidea*. Basal-rosetted, perennial succulent. H 12in (30cm), S indefinite. Has rigid, blue-gray leaves, each with spines up margins and a long, dark spine at tip. Flower stem, to 5ft (1.5m) long, bears yellow flowers in summer. Z11

A. victoriae-reginae (illus. p.482). Slow-growing, variable perennial succulent. H and S to 20in (50cm). Produces basal rosettes of tight-fitting, triangular to oblong, straight or incurved, dark green leaves, marked white, usually along the margins. Leaf margins are usually entire,

but may have small white spines. Flower stem, to 16ft (5m) long, bears a narrow, dense spike of funnel-shaped, creamy-white flowers, sometimes tinged purple, ¾–1¼in (2–3cm) long, in summer. Z9–11.

‘Compacta’ is slow-growing and smaller, H 12in (30cm), S 18in (45cm); deep green leaves have a small, terminal spine and spineless, white margins.

A. zebra. Basal-rosetted, perennial succulent. H 3ft (1m), S 1½–3ft (45–60cm). Lance-shaped, thick, rigid, rough, wavy-margined, patterned, light gray leaves have curved, gray spines. Flower stem, 20–25ft (6–8m) long, bears a narrow panicle of bell-shaped, yellow flowers, 1¼in (4.5cm) long, in summer. Z11

AGERATINA

ASTERACEAE/COMPOSITAE

Genus of evergreen shrubs, subshrubs and herbaceous perennials, grown mainly for their flowers, some also for their architectural foliage. Requires full light or partial shade. Will grow in any conditions, although most species prefer moist but well-drained soil. Water containerized plants freely when in full growth, moderately at other times. Prune shrubs lightly after flowering or in spring. Propagate by seed in spring; shrubs and subshrubs may also be propagated by softwood or greenwood cuttings in summer, perennials by division in early spring or fall. Red spider mite and whitefly may be troublesome.

A. altissima, syn. *Eupatorium ageratoides*, *E. rugosum*, *E. urticifolium* (Hardy age, Mist flower, White snakeroot). Erect perennial. H 5–6ft (1.5–1.8m), S 2ft (60cm). Produces nettlelike, gray-green leaves and, in late summer, bears dense, flat, white flower heads. Z4–8

A. ligustrina, syn. *Eupatorium ligustrinum*, *E. micranthum*, *E. weinmannianum*. Evergreen, rounded shrub. H and S 6–15ft (2–5m). Has elliptic to lance-shaped, bright green leaves and, in fall, fragrant, groundsel-like, white or pink flowers are produced in flattened clusters, 4–8in (10–20cm) across. Z13–15

AGERATUM

Floss flower

ASTERACEAE/COMPOSITAE

Genus of annuals and biennials, grown for their rounded clusters of flowers, which are attractive to butterflies. Useful for bedding. Grow in sun and fertile, well-drained soil, which should not be allowed to dry out or growth and flowering will be poor. Deadhead regularly for continuous flowering. Propagate by seed sown outdoors in late spring.

A. houstonianum cultivars. Moderately fast-growing, compact, hummock-forming annuals. H and S 6–12in (15–30cm). All have ovate, downy, mid-green leaves, with heart-shaped bases. Rounded, brushlike clusters of small flower heads are borne just above the foliage from midsummer until first frosts. Z10–11. ‘**Blue Danube**’ illus. p.313. ‘**Blue Mink**’ illus. p.314.

Hawaii Series are uniform, compact plants, with deep to pale blue or white flower heads. ‘**Pacific**’ has a neat habit